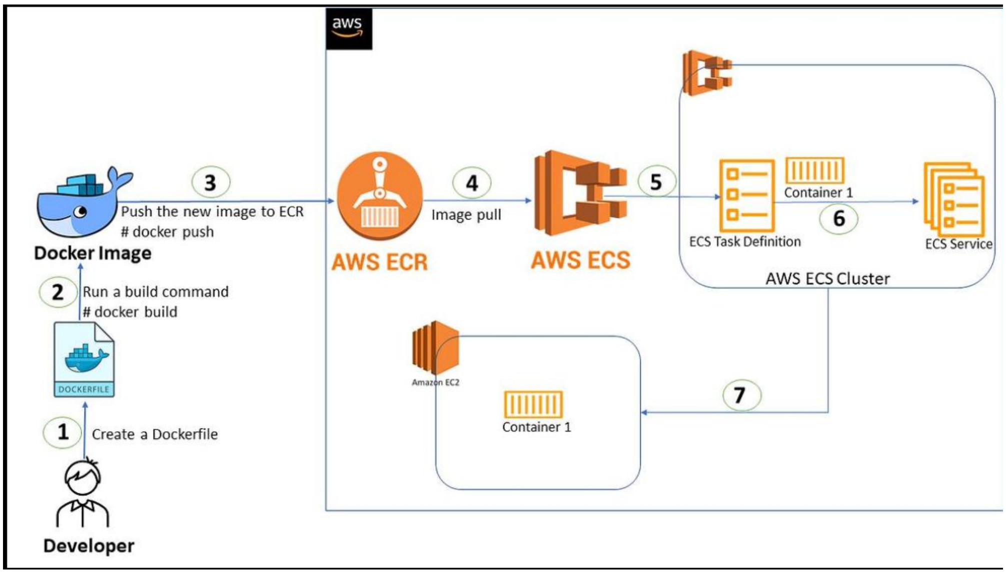
Name : Vivek Chaudhari

Project : 8

Docklet: Containerized Flask app deployed on AWS



Project Overview

Docklet is a cloud-based deployment project where a Python Flask web application is containerized using Docker and deployed on AWS. The Docker image is stored in a container registry and executed using a managed container orchestration service. This approach ensures consistent deployment, scalability, and eliminates environment dependency issues

Working Flow (End-to-End)

1. A Flask web application is developed using Python
2. The application is containerized using Docker (includes code + dependencies)
3. The Docker image is pushed to a container registry
4. The container orchestration service pulls the image
5. The application runs inside containers in the cloud
6. Users access the application via a public endpoint

⚙️ AWS Services Used & Their Role

📦 Amazon Elastic Container Registry

- Stores Docker images securely
- Acts as a central repository for container images
- Enables version control for deployments

⚙️ Amazon Elastic Container Service

- Deploys and manages containers
- Ensures application availability
- Supports scaling and load distribution

🐳 Core Technologies

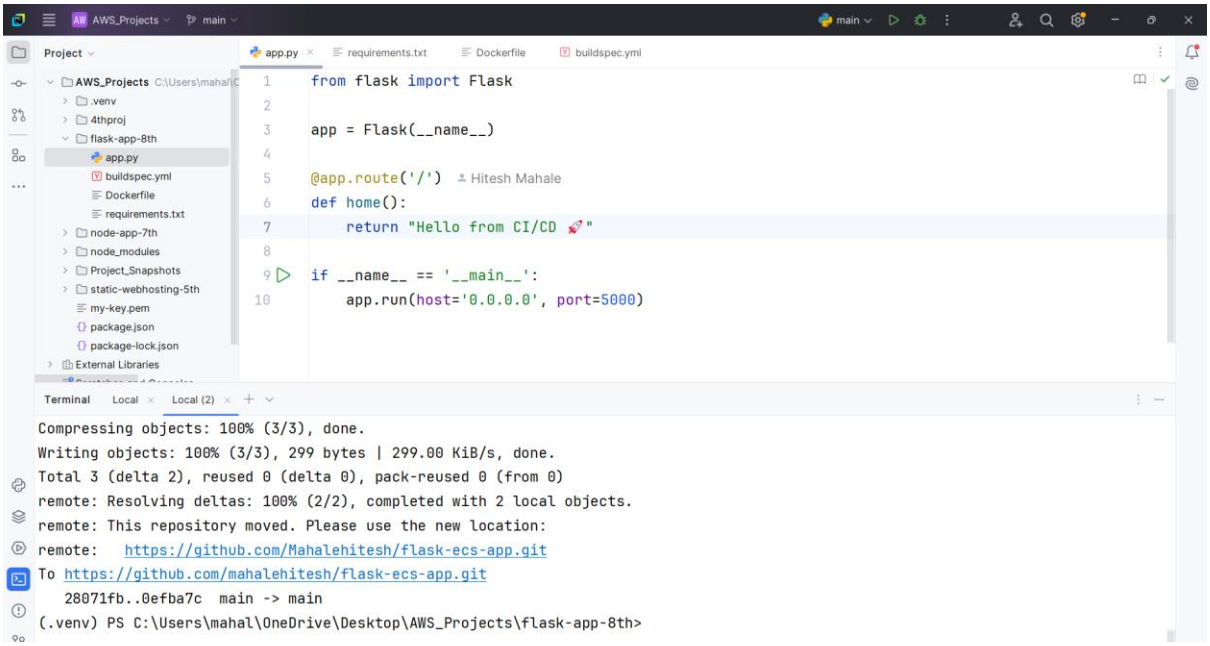
- **Flask** → Backend web application
- **Docker** → Containerization platform
- **AWS Cloud** → Deployment environment

🎯 Key Objectives Achieved

✓ Eliminated manual deployment issues
✓ Ensured consistent runtime environment
✓ Enabled scalable container-based architecture
✓ Followed modern DevOps practices

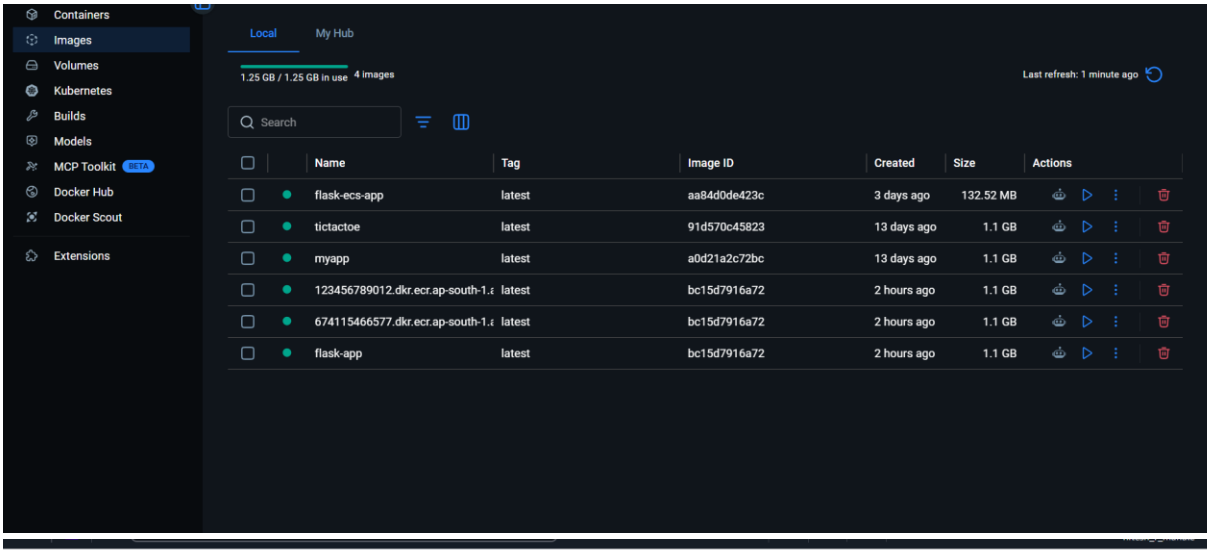
💼 Business / Real-World Value

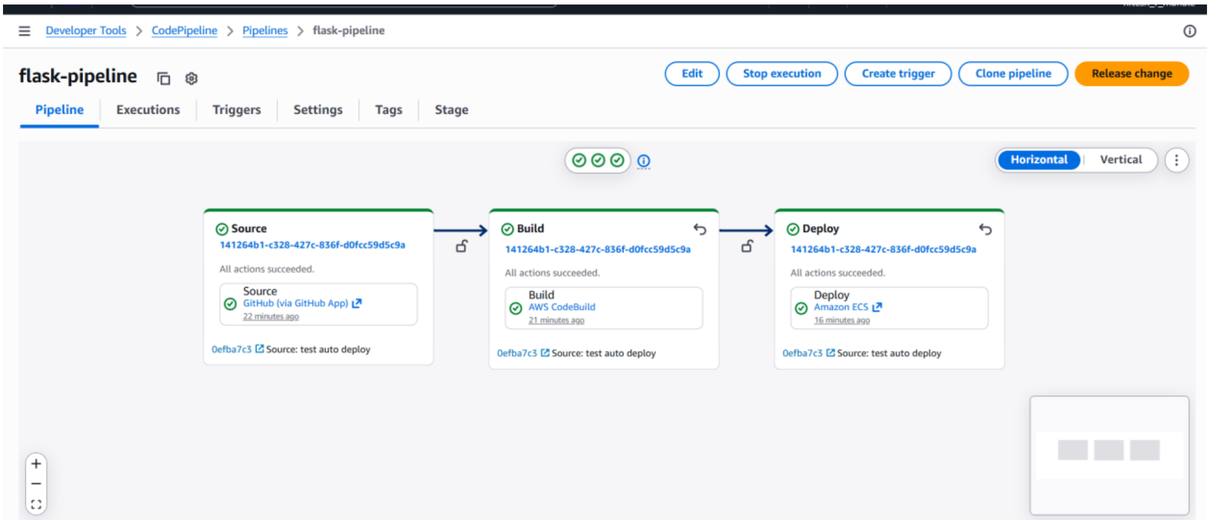
- Faster deployment cycles
- Easy scalability for growing traffic
- Reduced environment-related bugs
- Production-ready cloud architecture

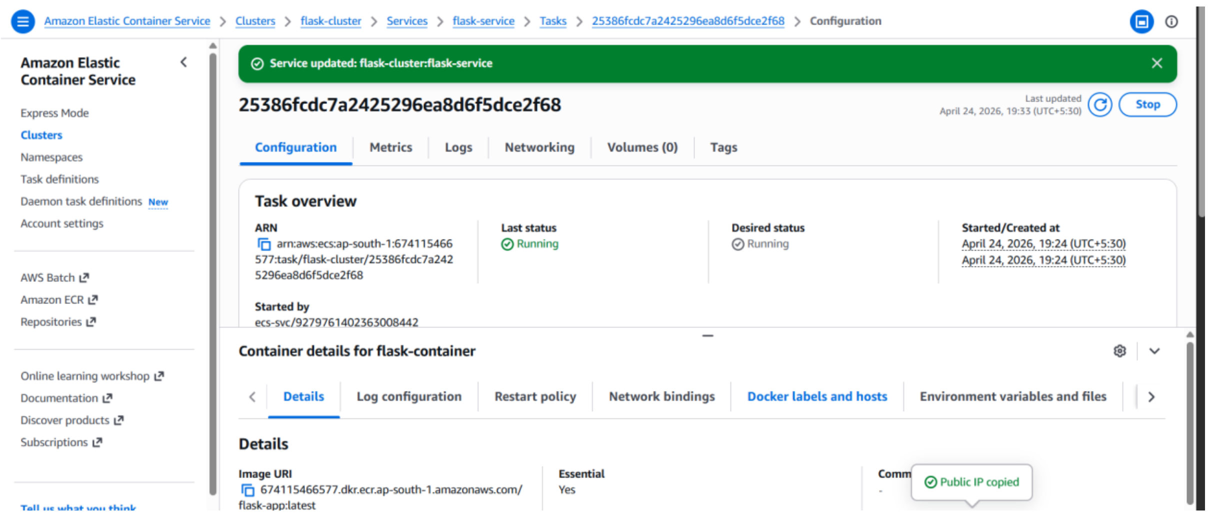


Hello from Docker Flask App 🚀

Hello from CI/CD 🚀


Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Local
My Hub
1.25 GB / 1.25 GB in use 4 images
Last refresh: 1 minute ago
Search
Name Tag Image ID Created Size Actions
flask-ecs-app latest aa84d0de423c 3 days ago 132.52 MB
tictactoe latest 91d570c45823 13 days ago 1.1 GB
myapp latest a0d21a2c72bc 13 days ago 1.1 GB
123456789012.dkr.ecr.ap-south-1.a latest bc15d7916a72 2 hours ago 1.1 GB
674115466577.dkr.ecr.ap-south-1.a latest bc15d7916a72 2 hours ago 1.1 GB
flask-app latest bc15d7916a72 2 hours ago 1.1 GB


Developer Tools > CodePipeline > Pipelines > flask-pipeline
flask-pipeline
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Stage
Horizontal
Vertical
Source
141264b1-c328-427c-836f-d0fcc59d5c9a
All actions succeeded.
Source
GitHub (via GitHub App)
22 minutes ago
0efba7c3 Source: test auto deploy
Build
141264b1-c328-427c-836f-d0fcc59d5c9a
All actions succeeded.
Build
AWS CodeBuild
21 minutes ago
0efba7c3 Source: test auto deploy
Deploy
141264b1-c328-427c-836f-d0fcc59d5c9a
All actions succeeded.
Deploy
Amazon ECS
16 minutes ago
0efba7c3 Source: test auto deploy


Amazon Elastic Container Service > Clusters > flask-cluster > Services > flask-service > Tasks > 25386fcdc7a2425296ea8d6f5dce2f68 > Configuration
Amazon Elastic Container Service
Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions New
Account settings
AWS Batch
Amazon ECR
Repositories
Online learning workshop
Documentation
Discover products
Subscriptions
Service updated: flask-cluster:flask-service
25386fcdc7a2425296ea8d6f5dce2f68
Last updated
April 24, 2026, 19:33 (UTC+5:30)
Stop
Configuration
Metrics
Logs
Networking
Volumes (0)
Tags
Task overview
ARN
arn:aws:ecs:ap-south-1:674115466577:task/flask-cluster/25386fcdc7a2425296ea8d6f5dce2f68
Last status
Running
Desired status
Running
Started/Created at
April 24, 2026, 19:24 (UTC+5:30)
April 24, 2026, 19:24 (UTC+5:30)
Started by
ecs-svc/9279761402363008442
Container details for flask-container
Details
Log configuration
Restart policy
Network bindings
Docker labels and hosts
Environment variables and files
Details
Image URI
674115466577.dkr.ecr.ap-south-1.amazonaws.com/flask-app:latest
Essential
Yes
Public IP copied